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SWITCHING CONVERTER CONTROLLER WITH ADAPTIVE PULSE-FREQUENCY MODULATION TIMER CIRCUIT

Abstract

A switching converter controller includes: mode control logic; and a pulse-frequency modulation (PFM) timer circuit coupled to the mode control logic. The PFM timer circuit includes: a first terminal; a second terminal; a digital-to-analog converter (DAC); a current source; a capacitor; and a comparator. A first terminal of the DAC is coupled to the first terminal of the PFM timer circuit. The current source has a first terminal, a second terminal, and a third terminal. The second terminal of the current source is coupled to a second terminal of the DAC. A first terminal of the capacitor is coupled to the first terminal of the current source. The comparator has first, second, and third terminals. The first terminal of the comparator is coupled to the first terminal of the capacitor. The third terminal of the comparator is coupled to the second terminal of the PFM timer circuit.

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Background/Summary

BACKGROUND

[0001] Switching converters are used to provide a direct-current (DC) output voltage (VOUT) based on an input voltage (VIN). A typical switching converter includes: a power stage with switches and an inductor; and a controller for the switches of the power stage. Switching converter efficiency varies depending on proper management of switching control responsive to variations in VIN, VOUT, load, operating frequency, and inductance. Mode transitions of the controller in response to load changes (e.g., light load to heavy load, or heavy load to light load) are problematic, often causing inefficient operation and/or VOUT ripple.

SUMMARY

[0002] In an example, a switching converter controller includes: mode control logic; and a pulse-frequency modulation (PFM) timer circuit coupled to the mode control logic. The PFM timer circuit includes: a first terminal; a second terminal; a digital-to-analog converter (DAC); a current source; a capacitor; and a comparator. The DAC has a first terminal and a second terminal. The first terminal of the DAC is coupled to the first terminal of the PFM timer circuit. The current source has a first terminal, a second terminal, and a third terminal. The second terminal of the current source is coupled to the second terminal of the DAC. The capacitor has a first terminal and a second terminal. The first terminal of the capacitor is coupled to the first terminal of the current source. The comparator has first, second, and third terminals. The first terminal of the comparator is coupled to the first terminal of the capacitor. The third terminal of the comparator is coupled to the second terminal of the PFM timer circuit.

[0003] In another example, a switching converter controller includes: mode control logic; and a PFM timer circuit coupled to the mode control logic. The PFM timer circuit is configured to: receive a digital control signal; adjust a current of the PFM timer circuit responsive to the digital control signal; charge a capacitor responsive to the adjusted current; compare a charge on the capacitor to a reference voltage to obtain a PFM timer result; and output the PFM timer result.

[0004] In yet another example, a system includes: a power stage; and a switching converter controller coupled to the power stage. The switching converter controller includes a PFM timer circuit and is configured to: identify a PFM mode trigger; provide adaptation control signals to the PFM timer circuit responsive to the PFM mode trigger; monitor adaptation results of the PFM timer circuit; and transition to a PFM mode responsive to the adaptation results of the PFM timer circuit indicating PFM timing is ready.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a diagram showing an example system.

[0006] FIGS. 2A to 2C are diagrams showing example waveforms and thresholds for different modes of a switching converter controller.

[0007] FIG. 3 is another diagram showing example waveforms and thresholds for different modes of a switching converter controller.

[0008] FIG. 4 is a diagram showing example components of a switching converter controller.

[0009] FIG. 5 is a diagram showing example mode control logic of a switching converter controller.

[0010] FIG. 6 is a timing diagram showing example waveforms and thresholds during transition from a pulse-width modulation (PWM) mode to a pulse-frequency modulation (PFM) mode.

[0011] FIG. 7 is a diagram showing example waveforms and thresholds for a switching converter

controller without an adaptive PFM timer.

[0012] FIG. **8** is a diagram showing example waveforms and thresholds for a switching converter controller with an adaptive PFM timer.

[0013] FIG. **9** is a flowchart showing an example switching converter controller method.

[0014] FIG. **10** is a flowchart showing an example PFM timer circuit method.

DETAILED DESCRIPTION

[0015] The same reference numbers or other reference designators are used in the drawings to designate the same or similar features. Such features may be the same or similar either by function and/or structure.

[0016] FIG. **1** is a diagram showing an example system **100**. The system **100** includes a power supply **102**, a power stage **106**, an output capacitor (COUT), a load **142**, and a controller **148**. The power supply **102** has a terminal **104**. The power stage **106** has a first terminal **108**, a second terminal **110**, a third terminal **112**, a fourth terminal **114**, a fifth terminal **116**, and a ground terminal **118**. COUT has a first terminal and a second terminal. The load **142** has a terminal **144** and a ground terminal **146**. The controller **148** has a first terminal **149**, a second terminal **150**, a third terminal **151**, a fourth terminal **152**, a fifth terminal **153**, and a ground terminal **154**.

[0017] As shown, the power stage **106** includes a high-side (HS) switch **120**, a low-side (LS) switch **128**, and an inductor **136** in the arrangement shown. In some examples, the HS switch **120**, LS switch **128**, and related drivers are components of an integrated circuit (IC), and the inductor **136** is an external component relative to the IC. The arrangement of components for the power stage **106** of FIG. **1** is referred to as a buck converter topology, where the output voltage (VOUT) is lower than the input voltage (VIN). In other examples, the topology of the power stage **106** may vary. In the example of FIG. **1**, the HS switch **120** has a first terminal **122**, a second terminal **124**, and a control terminal **126**. The LS switch **128** has a first terminal **130**, a second terminal **132**, and a control terminal **134**. In some examples, the HS switch **120** may be a p-channel field-effect transistor (“PFET”) or an n-channel field-effect transistor (“NFET”), and the LS switch **128** may be an NFET. The inductor **136** has a first terminal **138** and a second terminal **140**.

[0018] The controller **148** includes valley control circuitry **155**, peak control circuitry **158**, an adaptive pulse-frequency modulation (PFM) timer circuit **164**, mode control logic **170**, and driver circuitry **184**. The valley control circuitry **155** has input(s) **156** and an output **157**. The peak control circuitry **158** has input(s) **160** and an output **162**. The adaptive PFM timer circuit **164** has first terminal(s) **166** and a second terminal **168**. The mode control logic **170** has a first terminal **172**, a second terminal **174**, a third terminal **176**, a fourth terminal **178**, a fifth terminal **180**, and a sixth terminal **182**. The driver circuit **184** has a first terminal **186**, a second terminal **188**, a third terminal **190**, and a fourth terminal **192**.

[0019] The first terminal **108** of the power stage **106** is coupled to the terminal **104** of the power supply **102** and receives VIN. The first terminal **149** of the controller **148** is also coupled to the terminal **104** of the power supply **102** and/or the first terminal **108** of the power stage **106**. The second terminal **110** of the power stage **106** is coupled to the fourth terminal **152** of the controller **148** and receives a HS switch control signal (HS_CS). The third terminal **112** of the power stage **106** is coupled to the fifth terminal **153** of the controller **148** and receives an LS switch control signal (LS_CS). The fourth terminal **114** of the power stage **106** is coupled to the terminal **144** of the load **142** and provides VOUT. The first terminal of COUT is also coupled to the fourth terminal **114** of the power stage **106** and the terminal **144** of the load **142**. The second terminal of COUT is coupled to a ground terminal. The fifth terminal **116** of the power stage **106** is coupled to the second terminal **150** of the controller **148** and provides a switch node voltage (V.sub.SW) at switch node **135**. The ground terminal **118** of the power stage **106**, the ground terminal **154** of the controller **148**, and the ground terminal **146** of the load **142** are coupled to ground.

[0020] As shown, the first terminal **122** of the HS switch **120** is coupled to the first terminal **108** of the power stage **106**. The second terminal **124** of the HS switch **120** is coupled to the first terminal

130 of the LS switch **128** and to the first terminal **138** of the inductor **136**. The control terminal **126** of the HS switch **120** is coupled to the second terminal **110** of the power stage **106**. The second terminal **132** of the LS switch **128** is coupled to the ground terminal **118** of the power stage **106**. The control terminal **134** of the LS switch **128** is coupled to the third terminal **112** of the power stage **106**. As shown, the fifth terminal **116** of the power stage **106** is coupled second terminal **124** of the HS switch **120**, the first terminal **130** of the LS switch **128**, and the first terminal **138** of the inductor **136**. The second terminal **140** of the inductor **136** is coupled to the fourth terminal **114** of the power stage **106**. In the example of FIG. 1, the second terminal **124** of the HS switch **120**, the first terminal **130** of the LS switch **128**, and the first terminal **138** of the inductor **136** are coupled to a switch node **135** between the HS switch **120** and the LS switch **128**.

[0021] As shown, the input(s) **156** of the valley control circuitry **155** receive control signal(s) CS1. In some examples, CS1 includes a valley threshold and an inductor current sense signal. In some examples, the valley threshold and/or the inductor current sense signal are ramped. The input(s) **160** of the peak control circuitry **158** receive control signal(s) CS2. In some examples, CS2 includes a peak threshold and an inductor current sense signal. The first terminal(s) **166** of the adaptive PFM timer circuit **164** receive control signal(s) CS3. In some examples, CS3 includes a control voltage (e.g., V_CTRL herein), a digital control signal, and switch control signals. In some examples, V_CTRL is the error result between VOUT and a reference voltage (VREF). In some examples, the digital control signal and the switch control signals are provided by the mode control logic **170**. For example, the digital control signal and the switch control signals may be included in the control signals CS4 provided by the mode control logic **170**.

[0022] The first terminal **172** of the mode control logic **170** is coupled to the output **157** of the valley control circuitry **155**. The second terminal **174** of the mode control logic **170** is coupled to the output **162** of the peak control circuitry **158**. The third terminal **176** of the mode control logic **170** is coupled to the second terminal **168** of the adaptive PFM timer circuit **164**. The fifth terminal **180** of the mode control logic **170** is coupled to the first terminal **186** of the driver circuitry **184**. The sixth terminal **182** of the mode control logic **170** is coupled to the second terminal **188** of the driver circuitry **184**. The third terminal **190** of the driver circuitry **184** is coupled to the fourth terminal **152** of the controller **148**. The fourth terminal **192** of the driver circuitry **184** is coupled to the fifth terminal **153** of the controller **148**.

[0023] In operation, the controller **148** is configured to: receive VIN at its first terminal **149**; receive V.sub.SW at its second terminal **150**; receive VOUT at its third terminal **151**; provide HS_CS at its fourth terminal **152** responsive to VIN, V.sub.SW, VOUT, and the operations of the valley control circuitry **155**, the peak control circuitry **158**, the adaptive PFM timer circuit **164**, the mode control logic **170**, and the driver circuitry **184**; and provide LS_CS at its fifth terminal **153** responsive to VIN, V.sub.SW, VOUT, and the operations of the valley control circuitry **155**, the peak control circuitry **158**, the adaptive PFM timer circuit **164**, the mode control logic **170**, and the driver circuitry **184**. In some examples, example modes supported by the controller **148** include a pulse-width modulation (PWM) mode and a PFM mode. In some examples, transitions from the PWM mode to the PFM mode are based on the controller **148** is configured to: identify a PFM mode trigger; provide adaptation control signals (e.g., CS4 or related control signals) to the adaptive PFM timer circuit **164** responsive to the PFM mode trigger; monitor adaptation results of the adaptive PFM timer circuit **164** responsive to the adaptation control signals; and transition to a PFM mode responsive to the adaptation results indicating PFM timing is ready.

[0024] In operation, the power stage **106** is configured to: receive VIN at its first terminal **108**; receive HS_CS at its second terminal **110**; receive LS_CS at its third terminal **112**; provide VOUT at its fourth terminal **114** responsive to VIN, HS_CS, and LS_CS; provide V.sub.SW at its fifth terminal **116** responsive to VIN, HS_CS, and LS_CS; and provide a coupling between the ground terminal **118** and ground. More specifically, the HS switch **120** couples VIN to the switch node **135** responsive to HS_CS, which increases the current in the inductor **136**. The LS switch **128** couples

the ground terminal **118** to the switch node **135** responsive to LS_CS, which decreases the current in the inductor **136**. The average current in the inductor **136** is considered the load current provided to the load **142**. In some examples, VIN may be 2.5 V to 40 V, VOUT may be 0.3 V to 6 V. In some examples, I_out may be 0 to 1 A during PFM mode. During PWM mode, I_out may be 0 to 10 A. [0025] Example modes supported by the controller **148** include a pulse-width modulation (PWM) mode and a PFM mode. In some examples, transitions from PWM mode to PFM mode are based on: identifying a PFM mode trigger (e.g., a light-load condition or a low-power mode signal); provide adaptation control signals (e.g., CS4 or related control signals) to the adaptive PFM timer circuit **164** responsive to the PFM mode trigger; monitor adaptation results of the adaptive PFM timer circuit **164** responsive to the adaptation control signals; and transition to the PFM mode responsive to the adaptation results indicating PFM timing is aligned with a target PFM timing. In some examples, PFM timing is adapted so that the PFM timer period matches the PWM switching period for PFM mode entry. After PFM mode entry, PFM timing is adjusted as needed responsive to the load current. In some examples, the PFM mode may keep the last value of the multi-bit digital control signal for the adaptive PFM timer circuit **164** at the time of transition to PFM mode and use V_CTRL to adjust the PFM timing as needed. In some examples, the adaptive PFM timer circuit **164** includes a digital-to-analog converter (DAC) to generate a current source control signal responsive to the digital multi-bit digital control signal. The current source control signal is used to adjust an offset current of the adaptive PFM timer circuit **164**, which affects the rate at which a capacitor (e.g., C_pfm herein) is charged. In some examples, the multi-bit digital control signal has 6 bits (e.g., Sel_pfm_timer<5:0> herein).

[0026] FIGS. 2A to 2C are diagrams **200**, **210**, and **220** showing example waveforms and thresholds for different modes of a switching converter controller. In some examples, the switching converter controller is a current mode buck converter controller with a PFM timer.

[0027] In the diagram **200** of FIG. 2A, example waveforms and thresholds are shown for a PWM mode interval **202** and a PFM mode interval **204**. The waveforms and thresholds for the PWM mode interval **202** include inductor current (I.sub.L), output current (I_out), a valley current threshold (I.sub.VALLEY), a switching clock signal (Clk_sw), and a valley compare result (Valley_comp). During the PWM mode interval **202**, IL ramps up (e.g., during a HS interval in which the HS switch **120** turned on) until Clk_sw is asserted. IL subsequently ramps down (e.g., during an LS interval in which the LS switch **128** turned on) until reaching I.sub.VALLEY. Each time IL reaches I.sub.VALLEY, Valley_comp is asserted, which ends an LS interval and starts a subsequent HS interval. With the waveforms and thresholds of the PWM mode interval **202** an average I_out is maintained. As desired, control parameters related to IL may be adjusted to increase or decrease I_out.

[0028] The waveforms and thresholds for the PFM mode interval **204** include I.sub.L, I_out, a PFM timing signal (Start_pfm_pulse), and a PFM switching period (T_sw). During the PFM mode interval **204**, IL ramps up and down responsive to a peak limit detection, a peak limit timer, zero current detection, a PFM timer interval (e.g., T_sw herein), and/or a PFM pulse start signal (e.g., Start_pfm_pulse). In some examples, when IL reaches a peak current threshold (e.g., I.sub.PEAK herein) or when a timer threshold is reached, a low-side interval starts and causes IL to ramp down. When IL reaches a zero current threshold (e.g., I.sub.ZERO herein), a pause interval starts. The pause interval ends responsive to the PFM timer interval (e.g., T_sw herein) and/or a related control signal (e.g., Start_pfm_pulse herein), which triggers a high-side interval. During the high-side interval, IL ramps up. During the PFM mode interval **204**, the spacing between PFM pulses during the PFM mode interval **204** is controlled by T_sw. In some examples, T_sw may be adjusted by the PFM timer.

[0029] In the example of FIG. 2A, the PFM pulse peak may be based on the inductor current ripple in the PWM mode interval **202**. One option is to control the output current in the PWM mode interval **202** and the PFM mode interval **204** based on V_CTRL. With this option, V_CTRL may

control the valley current level in PWM mode interval **202** and T_{sw} in the PFM mode interval **204**. In the PFM mode interval **204**, the PFM pulse width and T_{sw} affects I_{out}. In some examples,

$$[00001] I_{out} = I_{valley} + \frac{I_{ripple}}{2}$$

for the PWM mode interval **202**, while

$$[00002] I_{out} = \frac{I_{peak}}{2} * \frac{T_{pulse_width}}{T_{sw}}$$

for the PFM mode interval **204**.

[0030] When transitioning to the PFM mode interval **204**, mismatch in I_{out} for the PWM mode interval **202** and the PFM mode interval **204** can cause VOUT undershoot, overshoot and/or mode bouncing. I_{out} mismatches for the PWM mode interval **202** and the PFM mode interval **204** may be caused by: use of V_CTRL for control of both modes; PFM timer offsets causing variations in T_{sub.sw}; and/or inductor value variance (e.g., inductors may have a tolerance of +/-30%) causing variations in the width of the PFM pulse.

[0031] In the diagram **210** of FIG. 2B, example waveforms and thresholds are shown for a PWM mode interval **212** and a PFM mode interval **214**. The waveforms and thresholds for the PWM mode interval **212** include I_{sub.L}, I_{out}, I_{sub.VALLEY}, and a peak current threshold (I_{sub.PEAK}). The waveforms and thresholds for the PFM mode interval **214** include I_{sub.L}, I_{out}, I_{sub.VALLEY}, I_{sub.PEAK}, Start_pfm_pulse, and T_{sw}. In the diagram **210**, a slow frequency, high inductance scenario is represented.

[0032] In the diagram **220** of FIG. 2C, example waveforms and thresholds are shown for a PWM mode interval **222** and a PFM mode interval **224**. The waveforms and thresholds for the PWM mode interval **222** include I_{sub.L}, I_{out}, I_{sub.VALLEY}, and I_{sub.PEAK}. The waveforms and thresholds for the PFM mode interval **214** include I_{sub.L}, I_{out}, I_{sub.VALLEY}, I_{sub.PEAK}, Start_pfm_pulse, and T_{sw}. In the diagram **220**, a fast frequency, low inductance scenario is represented. For the diagrams **210** and **220**, the same load, the same I_{sub.VALLEY} levels, and the same I_{sub.PEAK} levels are assumed. However, the PFM timer needs to operate with different values of T_{sw} for the PFM mode intervals **214** and **224** even through the I_{sub.PEAK} levels are equal. Also, the PFM pulse width for the PFM mode interval **214** may vary from the PFM pulse width for the PFM mode interval **224** depending on duty ratio, switching frequency, and inductance. Without PFM timer tuning for different scenarios as described herein (e.g., the slow frequency, high inductance scenario of the diagram **210** versus the fast frequency, low inductance scenario of the diagram **220**) VOUT may undershoot, may undershoot, and/or may cause mode bounce.

[0033] FIG. 3 is another diagram **300** showing example inductor current, thresholds, and control signals for different modes of a switching converter controller. In the diagram **300** of FIG. 3, example waveforms and thresholds are shown for a PWM mode **302** and a PFM mode **304**. The waveforms and thresholds for the PWM mode **302** include I_{sub.L}, I_{out}, I_{sub.VALLEY}, Clk_{sw}, and Valley_comp. The waveforms and thresholds for the PFM mode **304** include I_{sub.L}, I_{out}, a zero current threshold (I_{sub.ZERO}), Start_pfm_pulse, and T_{sub.sw}.

[0034] For the diagram **300**, an adaptive PFM timer (e.g., the adaptive PFM timer circuit **164** in FIG. 1) is used. In some examples, an adaptive PFM timer operates to adjust an offset current of the adaptive PFM timer responsive to a digital control signal. In some examples, the digital control signal is based on a PFM pulse width before entering the PFM mode **304**. In some examples, a voltage related to I_{sub.ZERO} (the voltage given as “V_valley_zero” herein) is muxed to an input of the adaptive PFM timer to adjust the offset current. In some examples, V_valley_zero generates the same I_{out} in the PWM mode **302** and the PFM mode **304**. In some examples, V_valley_zero may equal the V_CTRL value that results in I_ZERO level in the PWM mode **302**. In such examples, V_valley_zero may generate consecutive PFM pulses in the PFM mode **304**, where the PFM pulse size is adapted based on the PWM current ripple. In some examples, the digital control

signal to the adaptive PFM timer is adjusted so that the offset current of the adaptive PFM timer results in a PFM timing period (T_{sw}) that matches with the PWM switching period (the period of Valley_comp). In some examples, mode control logic may compare the PWM switching period to PFM timer period. Based on the comparison, the digital control signal may be adjusted every second switching cycle. If the PFM timer period is shorter than PWM switching period, the digital control signal is adjusted to increase the offset current of the adaptive PFM timer. If the PFM timer period is longer than PWM switching period, the digital control signal is adjusted to decrease the offset current which steps the DAC downwards to decrease the offset current of the adaptive PFM timer.

[0035] FIG. 4 is a diagram showing example components of a switching converter controller **148A**. The switching converter controller **148A** is an example of the controller **148** in FIG. 1. As shown, the switching converter controller **148A** includes an adaptive PFM timer circuit **164A**, an error amplifier **432**, a resistor **R1**, a capacitor **C1**, a voltage-to-current converter **440**, a peak comparator **446**, a valley comparator **454**, mode control logic **170A**, and the driver circuitry **184**. The adaptive PFM timer circuit **164A** is an example of the adaptive PFM timer circuit **164** in FIG. 1. In the example of FIG. 4, the error amplifier **432**, the voltage-to-current converter **440**, and the peak comparator **446** are example components of the peak control circuitry **158** in FIG. 1. The error amplifier **432**, the voltage-to-current converter **440**, and the valley comparator **454** are example components of the valley control circuitry **155** in FIG. 1.

[0036] The error amplifier **432** has a first (inverting or “-”) terminal **434**, a second (non-inverting or “+”) terminal **436**, and a third terminal **438**. The voltage-to-current converter **440** has a first terminal **442**, a second terminal **443**, and a third terminal **444**. The peak comparator **446** has a first (non-inverting or “+”) terminal **448**, a second (inverting or “-”) terminal **450**, and a third terminal **452**. The valley comparator **454** has a first (non-inverting or “+”) terminal **456**, a second (inverting or “-”) terminal **458**, and a third terminal **460**. The mode control logic **170A** has the first terminal **172**, the second terminal **174**, the third terminal **176**, fourth terminals **178A** to **178C**, the fifth terminal **180**, and the sixth terminal **182**. In the example of FIG. 4, each of the fourth terminals **178A** to **178C** is an example of the fourth terminal **178** in FIG. 1. The driver circuitry **184** has the first terminal **186**, the second terminal **188**, the third terminal **190**, and the fourth terminal **192** described in FIG. 1.

[0037] In the example of FIG. 4, the adaptive PFM timer circuit **164A** has first terminals **166A**, **166B**, **166C**, and **166D**, and the second terminal **168**. Each of the first terminals **166A**, **166B**, **166C**, and **166D** are examples of the first terminal(s) **166** in FIG. 1. As shown, the adaptive PFM timer circuit **164A** includes a digital-to-analog converter (DAC) **402**, a first switch **S1**, a first current source **412**, a second current source **418**, a second switch **S2**, a capacitor **C_pfm**, and a comparator **424**. The DAC **402** has a first terminal **404** and a second terminal **406**. The first switch **S1** has a first terminal **T1**, a second terminal **T2**, a third terminal **T3**, and a fourth terminal **T4**. The first current source **412** has a first terminal **414**, a second terminal **415**, and a third terminal **416**. The second current source **418** has a first terminal **420**, a second terminal **421**, and a third terminal **422**. The second switch **S2** has a first terminal **T1**, a second terminal **T2**, and a third terminal **T3**. The capacitor **C_pfm** has a first terminal and a second terminal. The comparator **424** has a first (e.g., a non-inverting or “+”) terminal **426**, a second (e.g., an inverting or “-”) terminal **428**, and a third terminal **430**.

[0038] In the example of FIG. 4, the first terminal **404** of the DAC **402** is coupled to the first terminal **166A** of the adaptive PFM timer circuit **164A**. The second terminal **406** of the DAC **402** is coupled to the second terminal **421** of the second current source **418**. The first terminal **T1** of the first switch **S1** is coupled to a zero valley reference source (not shown) and receives V_{valley_zero} . The second terminal **T2** of the first switch **S1** is coupled to the first terminal **166B** of the adaptive PFM timer circuit **164A**. The third terminal **T3** of the first switch **S1** is coupled to the second terminal **415** of the first current source **412**. The fourth terminal **T4** of the first switch **S1**

is coupled to the first terminal **166C** of the adaptive PFM timer circuit **164A**. The first terminal **414** of the first current source **412** is coupled to a source voltage (VDD) terminal. The third terminal **416** of the first current source **412** is coupled to the first terminal **420** of the second current source **418**. The third terminal **416** of the first current source **412** is also coupled to first terminal T1 of the second switch S2, to the first terminal of the capacitor C_pfm, and to the first terminal **426** of the comparator **424**. The second terminal T2 of the second switch S2 is coupled to ground or a ground terminal. The third terminal T3 of the second switch S2 is coupled to the first terminal **166D** of the adaptive PFM timer circuit **164A**. The second terminal of the capacitor C_pfm is coupled to ground or a ground terminal. The second terminal **428** of the comparator **424** is coupled to a reference voltage source (not shown) and receives a PFM reference voltage (V_ref_pfm). The third terminal **430** of the comparator **424** is coupled to the second terminal **168** of the adaptive PFM timer circuit **164A**.

[0039] The first terminal **434** of the error amplifier **432** is coupled to an output terminal of a power stage (e.g., the fourth terminal **114** in FIG. 1) and receives VOUT as a feedback voltage (V_fb). As another option, V_fb may be scaled version of VOUT (e.g., using a voltage divider). The second terminal **436** of the error amplifier **432** is coupled to a reference voltage source (not shown) and receives a reference voltage (V_ref). The third terminal **438** of the error amplifier **432** is coupled to the first terminal of the resistor R1 and to the first terminal **442** of the voltage-to-current converter **440**. The second terminal of the resistor R1 is coupled to the first terminal of the capacitor C1. The second terminal of the capacitor C1 is coupled to ground or a ground terminal. The second terminal **443** of the voltage-to-current converter **440** is coupled to the second terminal **450** of the peak comparator **446**. The first terminal **448** of the peak comparator **446** is coupled to a sense current (I_sense) source. An example I_sense source is the fifth terminal **116** of the power stage **106** in FIG. 1 or related sense circuitry configured to monitor V_sub.SW or a related current. The third terminal **444** of the voltage-to-current converter **440** is coupled to the first terminal **456** of the valley comparator **454**. The second terminal **458** of the valley comparator **454** is coupled to the I_sense source (not shown).

[0040] The first terminal **172** of the mode control logic **170A** is coupled to third terminal **460** of the valley comparator **454**. The second terminal **174** of the mode control logic **170A** is coupled to the third terminal **452** of the peak comparator **446**. The third terminal **176** of the mode control logic **170A** is coupled to the second terminal **168** of the adaptive PFM timer circuit **164A**. The fourth terminal **178A** of the mode control logic **170A** is coupled to the first terminal **166A** of the adaptive PFM timer circuit **164A**. The fourth terminal **178B** of the mode control logic **170A** is coupled to the first terminal **166C** of the adaptive PFM timer circuit **164A**. The first terminal **166B** of the adaptive PFM timer circuit **164A** is coupled to the third terminal **438** of the error amplifier **432** and receives a control voltage (V_CTRL). The fourth terminal **178C** of the mode control logic **170A** is coupled to the first terminal **166D** of the adaptive PFM timer circuit **164A**. The fifth terminal **180** of the mode control logic **170A** is coupled to the first terminal **186** of the driver circuitry **184**. The sixth terminal **182** of the mode control logic **170A** is coupled to the second terminal **188** of the driver circuitry **184**. The third terminal **190** of the driver circuitry **184** is coupled to a control terminal of a first switch (e.g., the high-side switch HS_SW in FIG. 1). The fourth terminal **192** of the driver circuitry **184** is coupled to a control terminal of a second switch (e.g., the low-side switch LS_SW in FIG. 1).

[0041] The error amplifier **432** is configured to: receive V_fb at the first terminal **434**; receive V_ref at the second terminal **436**; and provide V_CTRL at the third terminal **438** responsive to V_fb and V_ref. The value of V_CTRL is based on the difference between V_fb and V_ref as well as the value of the resistor R1 and the value of the capacitor C1, which form compensation circuitry for the control loop. Specifically, R1 and C1 convert error amplifier current to voltage based on proportional integrator (PI) control. The transconductance of the error amplifier **432** defines the gain applied when converting the voltage error at the input of the error amplifier **432** to an output

current. The voltage-to-current converter **440** is configured to: receive V_CTRL at the first terminal **442**; provide a peak reference current (I_ref_peak) at the second terminal **443** responsive to V_CTRL; and provide a valley reference current (I_ref_valley) at the third terminal **444** responsive to V_CTRL. In some examples, voltage-to-current converter **440** converts V_CTRL to I_ref_valley based on a given transconductance in the PWM mode. If the PFM mode, the voltage-to-current converter **440** sets I_ref_valley to zero. In some examples, the voltage-to-current converter **440** may convert V_CTRL to I_ref_peak using the given transconductance. In some examples, the voltage-to-current converter **440** I_ref_valley and/or I_ref_peak generation is also based on a hysteresis current, which defines the difference between the valley and peak levels. In the PWM mode, the hysteresis current may be a predefined value (e.g., relative to a fixed current level). In the PFM mode, the hysteresis current may be relative to the inductor current ripple in the PWM mode.

[0042] The peak comparator **446** is configured to: receive I_sense at the first terminal **448**; receive I_ref_peak at the second terminal **450**; and provide a peak compare result (Peak_comp) at the third terminal **452** responsive to I_sense and I_ref_peak. The valley comparator **454** is configured to: receive I_ref_peak at the first (non-inverting) terminal **456**; receive I_sense at the second (inverting) terminal **458**; and provide a comparison result Valley_comp at the third terminal **460** responsive to I_sense and I_ref_valley.

[0043] The adaptive PFM timer circuit **164A** is configured to: receive a digital control signal Sel_pfm_timer<5:0> at the first terminal **166A**; receive V_CTRL at the first terminal **166B**; receive a PFM mode signal (PFM_mode) at the first terminal **166C**; receive a reset signal (Reset_pfm_timer) at the first terminal **166D**; and provide a PFM timer result (PFM_timer_comp) at the second terminal **168** responsive to Sel_pfm_timer<5:0>, V_CTRL, PFM_mode, and Reset_pfm_timer. In some examples, Start_pfm_pulse in FIGS. 2A, 2B, and 2C may be based on PFM_timer_comp being aligned with a target PFM timing as described herein. In some examples, Start_pfm_pulse may be delayed relative to PFM_timer_comp. For example, once PFM_timer_comp triggers, Start_pfm_pulse may not trigger until a next low-side interval ends or pause interval begins.

[0044] The mode control logic **170A** is configured to: receive Valley_comp at the first terminal **172**; receive Peak_comp at the second terminal **174**; receive PFM_timer_comp at the third terminal **176**; provide Sel_pfm_timer<5:0> at the fourth terminal **178A** to perform offset current adaptation operations for the adaptive PFM timer circuit **164A** responsive to a light load condition; provide PFM_mode at the fourth terminal **178B** responsive to adaptation operations of the adaptive PFM timer circuit **164A** being complete to indicate the PFM mode has started; provide Reset_pfm_timer at the fourth terminal **178C** during the offset current adaptation operations (e.g., every other PWM_CS cycle); control a PWM control signal (PWM_CS) state at the fifth terminal **180** responsive to Valley_comp, Peak_comp, a clock signal (e.g., Clk_sw in FIG. 5), PFM_timer_comp, and/or a current mode; and provide a high impedance (HIZ) control signal (HIZ_CS) at the sixth terminal **182** responsive to Valley_comp, Peak_comp, PFM_timer_comp and/or the current mode (e.g., PWM mode or PFM mode).

[0045] In some examples, the mode control logic **170A** is configured to control the state of PWM_CS at the fifth terminal **180** and the state of HIZ_CS at the sixth terminal **182** during PWM mode operations and during PFM mode operations of the switching converter controller **148A**. In some examples, during a PWM mode, the mode control logic **170A** is configured to maintain HIZ_CS in a second state (e.g., a de-asserted or logical low state). With HIZ_CS in the second state, PWM_CS controls the state of HS_CS and LS_CS during the PWM mode to support HS intervals and LS intervals. During the PWM mode, PWM_CS may be asserted responsive to Valley_comp being asserted (indicating the current of the inductor **136** has reached a valley threshold) and may be de-asserted responsive to a timer or clock signal (e.g., Clk_sw in FIG. 5) or responsive to Peak_comp being asserted (indicating the current of the inductor **136** has reached a

peak threshold). For each HS interval, HS_CS may be asserted while LS_CS is de-asserted. For each LS interval, HS_CS may be de-asserted while LS_CS is asserted. The duration of the HS interval relative to the LS interval for each cycle defines a duty cycle provided by the switching converter controller **148A**. For example, if the HS interval and the LS interval are equal in duration, the duty cycle provided by the switching converter controller **148A** is 50%.

[0046] In response to a light load condition, the mode control logic **170A** is configured to transition from the PWM mode to a PFM mode. To initiate transition from the PWM mode to the PFM mode, a light load condition is detected or a low-power mode control signal is received. During the transition interval, the mode control logic **170A** is configured to provide and adjust Sel_pfm_timer<5:0> at the fourth terminal **178** to adjust the offset current of the adaptive PFM timer circuit **164A** until assertion of PFM_timer_comp is aligned with the timing of PWM_CS, indicating the adaptive PFM timer circuit **164A** is ready for the PFM mode. As part of the offset current adjustment or adaptation, the mode control logic **170A** may assert Reset_pfm_timer every other cycle of PWM_CS. Once offset current adjustment or adaptation is complete (e.g., when PFM_timer_comp is aligned with a target), the mode control logic **170A** asserts PFM_mode at the fourth terminal **178B** and PFM mode operations begin. Example PFM mode operations include controlling PFM pulse timing based on V_CTRL and last value for Sel_pfm_timer<5:0> used during adaptation, controlling timing of HS intervals and LS intervals (ramp up time and ramp down time of I.sub.L), and asserting HIZ_CS as needed to provide a pause between HS intervals and LS intervals during the PFM mode.

[0047] Responsive to detecting a light load condition, the mode control logic **170** initiates a transition from the PWM to a PFM mode. In some examples, the light load condition may be detected using a comparator (not shown) to detect when V_CTRL drops below V_valley_zero (indicating the valley current level is below zero). As another option, a host system may provide a control signal (e.g., a low-power mode control signal) that triggers a transition from the PWM mode to the PFM mode. During the transition interval, the adaptive PFM timer circuit **164A** is configured to: receive Sel_pfm_timer<5:0> at the first terminal **166A**; receive PFM_mode at the first terminal **166C**; provide a current used to charge the capacitor C_pfm of the adaptive PFM timer circuit **164A** responsive to a zero valley reference voltage (V_valley_zero); adjust the current responsive to Sel_pfm_timer<5:0> and a related offset current; and assert PFM_timer_comp at the second terminal **168** responsive to the charge level (V_timer_ramp) of the capacitor C_pfm becoming greater than V_ref_pfm.

[0048] More specifically, during the PWM mode (e.g., PFM_mode de-asserted) the first switch S1 is controlled so that V_valley_zero is provided to the first current source **412**. The first current source **412** provides a current to charge the capacitor C_pfm based on V_valley_zero. The DAC **402** provides a control signal to the second current source **418** responsive to Sel_pfm_timer<5:0>. The second current source **418** provides an offset current responsive to the control signal from the DAC **402**. The offset current provided by the second current source **418** adjusts the current provided by the first current source **412**. The adjusted current (I_pfm_ramp) is used to charge the capacitor C_pfm. Once the charge level (V_timer_ramp) of the capacitor C_pfm becomes greater than V_ref_pfm, the comparator **424** asserts PFM_timer_comp at the second terminal **168**.

[0049] Once PFM_timer_comp is aligned with a target, the mode control logic **170A** de-asserts pfm_entry_adaptation and asserts PFM_mode to start PFM mode operations. During the PFM mode (e.g., PFM_mode asserted), the first switch S1 is controlled so that V_CTRL is provided to the first current source **412**. Thus, during the PFM mode, the first current source **412** provides a current to charge the capacitor C_pfm based on V_CTRL. In some examples, during the PFM mode, Sel_pfm_timer<5:0> may maintain the adaptation result value (setting the offset current to a fixed value), while V_CTRL varies the current from the first current source **412** to charge the capacitor C_pfm. Once the charge level (V_timer_ramp) of the capacitor C_pfm becomes greater than V_ref_pfm, the comparator **424** asserts PFM_timer_comp at the second terminal **168**. When

PFM_timer_comp is asserted, a next PFM pulse of the PFM mode is triggered. After PFM_timer_comp is asserted, the second switch S2 may reset the charge on the capacitor C_pfm responsive to Reset_pfm_timer.

[0050] In some examples, the adaptive PFM timer circuit **164A** selects between V_valley_zero and V_CTR using the first switch S1. With V_valley_zero, the PFM mode is initiated with same output current as the PWM mode. In some examples, V_valley_zero equals V_CTRL when there is a zero valley current level in the PWM mode. V_valley_zero may generate consecutive PFM pulses in PFM mode when the PFM pulse size is adapted based on the PWM current ripple. The mode control logic **160A** provides a digital signal (Sel_pfm_timer<5:0>) to select the offset current used to adjust the current of the adaptive PFM timer circuit **164A**. Again, the offset current is used to adjust I_pfm_ramp until the PFM timer period provided by the adaptive PFM timer circuit **164A** is aligned with the PWM switching period.

[0051] In some examples, the mode control logic **170A** is configured to compare the PWM switching period to the PFM timer period and adjust the digital signal to the DAC **402** every second switching cycle. If the PFM timer period is shorter than PWM switching period, the digital signal increases the output of the DAC **402** to increase the offset current. With an increased offset current, I_pfm_ramp is reduced, charging of the capacitor C_pfm is slower, and the PFM timer period increases. If the PFM timer period is longer than the PWM switching period, the digital signal decreases the output of the DAC **402** to decrease the offset current. With a decreased offset current, I_pfm_ramp is increased, charging of the capacitor C_pfm is faster, and the PFM timer period decreases.

[0052] During a PFM mode, the mode control logic **170A** is configured to assert HIZ_CS as needed between PFM pulses to enable pause intervals (e.g., both of the HS switch **120** and the LS switch **128** are turned off during each pause interval) between HS intervals and LS intervals. Also, during the PFM mode, PFM_pulse_comp may be asserted to initiate a PFM pulse (a HS interval) when HIZ_CS is de-asserted and PWM_CS is asserted. During the PFM mode, Peak_comp being asserted may initiate an LS interval when PWM_CS is de-asserted. During the PFM mode, Valley_comp may be used to detect zero current and trigger assertion of HIZ_CS.

[0053] The driver circuitry **184** is configured to receive PWM_CS at the first terminal **186**; receive HIZ_CS at the second terminal **188**; provide HS_CS with a first state (e.g., HS_CS asserted) at the third terminal **190** and LS_CS with a second state (e.g., LS_CS de-asserted) at the fourth terminal **192** responsive to PWM_CS having a first state (e.g., PWM_CS asserted) and HIZ_CS having a second state (e.g., HIZ_CS de-asserted); provide HS_CS with a second state (e.g., HS_CS de-asserted) at the third terminal **190** and LS_CS with a first state (e.g., LS_CS asserted) at the fourth terminal **192** responsive to PWM_CS having a second state (e.g., PWM_CS de-asserted) and HIZ_CS having a second state (e.g., HIZ_CS de-asserted); and provide HS_CS with a second state (e.g., HS_CS de-asserted) at the third terminal **190** and LS_CS with a second state (e.g., LS_CS de-asserted) at the fourth terminal **192** responsive to HIZ_CS having a first state (e.g., HIZ_CS asserted).

[0054] FIG. 5 is a diagram showing example mode control logic **170B** of a switching converter controller (e.g., the controller **148** of FIG. 1, or the switching converter controller **148A** in FIG. 4). The mode control logic **170B** is an example of the mode control logic **170** in FIG. 1, or the mode control logic **170A** in FIG. 4. As shown, the mode control logic **170B** has the first terminal **172**, the second terminal **174**, the third terminal **176**, the fourth terminal **178A**, the fourth terminal **178B**, the fourth terminal **178C**, the fifth terminal **180**, the sixth terminal **182**, and a seventh terminal **502**. The mode control logic **170B** includes switch control circuitry **504**, PFM mode entry circuitry **520**, and PFM control circuitry **528**. The switch control circuitry **504** has a first terminal **506**, a second terminal **508**, a third terminal **510**, a fourth terminal **512**, a fifth terminal **514**, a sixth terminal **516**, and a seventh terminal **518**. The PFM mode entry circuitry **520** has a first terminal **521**, a second terminal **522**, a third terminal **524**, and a fourth terminal **526**. The PFM control circuitry **528** has a

first terminal **530**, a second terminal **532**, a third terminal **534**, a fourth terminal **536**, a fifth terminal **538**, and a sixth terminal **540**.

[0055] In the example of FIG. 5, the first terminal **506** of the switch control circuitry **504** is coupled to the seventh terminal **502** of the mode control logic **170B**. The second terminal **508** of the switch control circuitry **504** is coupled to the first terminal **172** of the mode control logic **170B**. The third terminal **510** of the switch control circuitry **504** is coupled to the third terminal **176** of the mode control logic **170B**. The fourth terminal **512** of the switch control circuitry **504** is coupled to the second terminal **174** of the mode control logic **170B**. The fifth terminal **514** of the switch control circuitry **504** is coupled to the third terminal **524** of the PFM mode entry circuitry **520**, the third terminal **534** of the PFM control circuitry **528**, and to the fourth terminal **178B** of the mode control logic **170B**. The sixth terminal **516** of the switch control circuitry **504** is coupled to the fifth terminal **180** of the mode control logic **170B**. The seventh terminal **518** of the switch control circuitry **504** is coupled to the sixth terminal **182** of the mode control logic **170B**.

[0056] The first terminal **521** of the PFM mode entry circuitry **520** receives a light load condition signal (LIGHT_LOAD). In some examples, the state of LIGHT_LOAD is determined by detecting when IL or average IL drops below a threshold. As another option, a low-power request or trigger may be used to assert LIGHT_LOAD. The second terminal **522** of the PFM mode entry circuitry **520** is coupled to the sixth terminal **540** of the PFM control circuitry **528**. The fourth terminal **526** is coupled to the second terminal **532** of the PFM control circuitry **528**. The first terminal **530** of the PFM control circuitry **528** is coupled to the sixth terminal **516** of the switch control circuitry **504**. The fourth terminal **536** of the PFM control circuitry **528** is coupled to the fourth terminal **178A** of the mode control logic **170B**. The fifth terminal **538** of the PFM control circuitry **528** is coupled to the fourth terminal **178C** of the mode control logic **170B**.

[0057] The PFM mode entry circuitry **520** is configured to: receive LIGHT_LOAD at the first terminal **521**; receive a PFM adaptation complete signal (PFM_adaptation_done) at the second terminal **522**; control the state of PFM_mode at the third terminal **524** responsive to LIGHT_LOAD and PFM_adaptation_done; control the state of a PFM adaptation control signal (PFM_entry_adaptation) at the fourth terminal **526** responsive to LIGHT_LOAD and PFM_adaptation_done. In some examples, PFM_entry_adaptation are asserted responsive to LIGHT_LOAD being asserted. PFM_entry_adaptation stays asserted during the transition interval until PFM_adaptation_done is asserted. Responsive to PFM_adaptation_done being asserted, indicating the PFM timer adaptation is complete and thus the transition interval is complete, the PFM mode entry circuitry **520** is configured to assert PFM_mode and de-assert PFM_entry_adaptation.

[0058] In some examples, the PFM control circuitry **528** is configured to adjust Sel_pfm_timer<5:0> at the fourth terminal **536** as needed while PFM_entry_adaptation is asserted to perform offset current adaptation operations of the adaptive PFM timer circuit **164A**. During offset current adaptation, the PFM control circuitry **528** is configured to receive PWM_CS at the first terminal **530**; and de-assert Reset_pfm_timer every second switching cycle based on PWM_CS (e.g., every other rising edge of PWM_CS). With Reset_pfm_timer de-asserted, the adaptive PFM timer circuit **164A** charges the capacitor C_pfm and generates V_timer_ramp. Once V_timer_ramp reaches a reference level (V_ref_pfm), the comparator **424** asserts PFM_timer_comp before PWM_CS rises (the adaptive PFM timer circuit **164A** operates faster than the PWM_CS period). During offset current adaptation, the PFM control circuitry **528** is configured to step up Sel_pfm_timer<5:0> to increase to the offset current, which decreases the current used to charge C_pfm (increasing the amount of time taken for V_timer_ramp to reach V_ref_pfm). If the comparator **424** does not trigger before PWM_CS rises, then the adaptive PFM timer circuit **164A** operates slower than the PWM_CS period. In such case, the PFM control circuitry continues to step up Sel_pfm_timer<5:0> as needed until the adaptive PFM timer circuit **164A** operates faster than the PWM_CS period. In some examples, offset current adaptation

continues by adjusting Sel_pfm_timer<5:0> up and down as needed until the timing of the adaptive PFM timer circuit **164A** is confirmed to closely approximate the timing of the PWM_CS period. Once the adaptation process is complete, the PFM control circuitry **528** is configured to assert PFM_adaptation_done, which causes the PFM mode entry circuitry **520** to end the transition or adaptation interval (e.g., by de-asserting PFM_entry_adaptation and asserting PFM_mode) and begin PFM mode operations.

[0059] In the example of FIG. 5, the switch control circuitry **504** is configured to: receive a clock signal (Clk_sw) at the first terminal **506**; receive Valley_comp at the second terminal **508**; receive PFM_pulse_comp at the third terminal **510**; receive Peak_comp at the fourth terminal **512**; receive PFM_mode at the fifth terminal **514**; control a state of PWM_CS at the sixth terminal **516** responsive to Clk_sw, Valley_comp, PFM_pulse_comp, Peak_comp, and PFM_mode; and control a state of HIZ_CS at the seventh terminal **518** responsive to Clk_sw, Valley_comp, PFM_pulse_comp, Peak_comp, and PFM_mode.

[0060] In some examples, during PWM mode operations, the switch control circuitry **504** is configured to maintain HIZ_CS in a second state (e.g., a de-asserted or logical low state). With HIZ_CS in the second state, PWM_CS controls the state of HS_CS and LS_CS during the PWM mode to support HS intervals and LS intervals. During the PWM mode, the switch control circuitry **504** may assert PWM_CS responsive to Valley_comp being asserted (indicating the current of the inductor **136** has reached a valley threshold) and may de-assert PWM_CS responsive to Clk_sw or responsive to Peak_comp being asserted (indicating the current of the inductor **136** has reached a peak threshold).

[0061] In some examples, the switch control circuitry **504** is configured to perform PWM mode operations during the transition interval. As another option, the switch control circuitry **504** may adjust some control options. For example, during the transition interval, the switch control circuitry **504** may use the peak comparator to trigger a low-side interval and may adjust the hysteresis current of I_ref_peak based on the PWM ripple.

[0062] During PFM mode operations, the switch control circuitry **504** is configured to assert HIZ_CS as needed to provide a pause between HS intervals and LS intervals. Also, during the PFM mode, the switch control circuitry **504** is configured to de-assert PWM_CS responsive to Peak_comp being asserted to initiate a LS interval; and assert HIZ_CS responsive to Valley_comp being asserted, indicating zero current detection.

[0063] FIG. 6 is a timing diagram **600** showing example inductor current, thresholds, and control signals during transition from a PWM mode to a PFM mode. For the timing diagram **600**, light load detection may be assumed to have already occurred. The waveforms represented in the timing diagram **600** include inductor current (I.sub.L), Valley_comp, PWM_CS, Reset_pfm_timer, I_pfm_ramp, V_timer_ramp, PFM_timer_comp, and PFM_adaptation_done. The timing diagram **600** also includes V_ref_pfm.

[0064] In the timing diagram **600**, IL ramps up and down and triggers Valley_comp whenever IL reaches a valley threshold (e.g., zero current). Each positive slope ramp of IL corresponds to a HS interval, in which the HS switch **120** is on and the LS switch **128** is off. Each negative slope ramp of IL corresponds to an LS interval, in which the HS switch **120** is off and the LS switch **128** is on. As shown, Reset_pfm_timer has a period that is twice the period of PWM_CS. At time t0, I_pfm_ramp has an initial value and Reset_pfm_timer is de-asserted, which results in V_timer_ramp beginning to rise starting at time t0. In some examples, the initial value for I_pfm_ramp may be based on a middle code (e.g., 6'b100000), which is the reset value of related digital flipflops. At time t1, V_timer_ramp crosses V_ref_pfm, resulting in assertion of PFM_timer_comp. Because PFM_timer_comp is asserted before the next rising edge of PWM_CS, offset current adaption is performed to increase the PFM_timer_comp period. At time t2, Reset_pfm_timer is asserted, which resets (e.g., grounds) V_timer_ramp. Also, I_pfm_ramp is reduced at time t2 (e.g., by increasing the offset current based on Sel_pfm_timer).

[0065] At time **t3**, $I_{\text{pfm_ramp}}$ has a reduced value relative to time **t0** and Reset_pfm_timer is de-asserted, which results in $V_{\text{timer_ramp}}$ beginning to rise starting at time **t3**. In some examples, the step size for $I_{\text{pfm_ramp}}$ depends on the DAC size and the maximum available current that the first current source **412** is able to provide in the PFM mode. In some examples, the maximum available current from the first current source **412** may be equal to $g_{\text{m_pfm}} \cdot V_{\text{valley_zero}}$. The maximum offset current (e.g., when a maximum DAC code is used) is selected to be smaller than the maximum available current from the first current source **412**. Without limitation, an example maximum available current from the first current source **412** may be 1 μA , a maximum offset current may be 945 nA, and a step size for $I_{\text{pfm_ramp}}$ may be 15 nA. In different examples, the step size may be adjusted to set accuracy or adaptation speed of PFM timer adaptation. In some examples, the step size of the least significant bit (LSB) of the DAC **402** may be used to select the step size for $I_{\text{pfm_ramp}}$.

[0066] At time **t4**, $V_{\text{timer_ramp}}$ crosses $V_{\text{ref_pfm}}$, resulting in assertion of PFM_timer_comp . Because PFM_timer_comp is asserted before the next rising edge of PWM_CS , offset current adaption will continue to increase the PFM_timer_comp period. At time **t5**, Reset_pfm_timer is asserted, which resets (e.g., grounds) $V_{\text{timer_ramp}}$. Also, $I_{\text{pfm_ramp}}$ is reduced at time **t5** (e.g., by increasing the offset current based on Sel_pfm_timer).

[0067] At time **t6**, $I_{\text{pfm_ramp}}$ has a reduced value relative to time **t3** and Reset_pfm_timer is de-asserted, which results in $V_{\text{timer_ramp}}$ beginning to rise starting at time **t6**. At time **t7**, $V_{\text{timer_ramp}}$ crosses $V_{\text{ref_pfm}}$, resulting in assertion of PFM_timer_comp . Because PFM_timer_comp is asserted before the next rising edge of PWM_CS , offset current adaption will continue to increase the PFM_timer_comp period. At time **t8**, Reset_pfm_timer is asserted, which resets (e.g., grounds) $V_{\text{timer_ramp}}$. Also, $I_{\text{pfm_ramp}}$ is reduced at time **t8** (e.g., by increasing the offset current based on Sel_pfm_timer).

[0068] At time **t9**, $I_{\text{pfm_ramp}}$ has a reduced value relative to time **t6** and Reset_pfm_timer is de-asserted, which results in $V_{\text{timer_ramp}}$ beginning to rise starting at time **t9**. At time **t10**, the next rising edge of PWM_CS is asserted before $V_{\text{timer_ramp}}$ crosses $V_{\text{ref_pfm}}$ (PFM_timer_comp is thus not asserted). Because the next rising edge of PWM_CS occurs before $V_{\text{timer_ramp}}$ crosses $V_{\text{ref_pfm}}$, offset current adaption will continue by decreasing the PFM_timer_comp period. At time **t10**, Reset_pfm_timer is asserted, which resets (e.g., grounds) $V_{\text{timer_ramp}}$. Also, $I_{\text{pfm_ramp}}$ is increased at time **t10** (e.g., by decreasing the offset current based on Sel_pfm_timer).

[0069] At time **t11**, $I_{\text{pfm_ramp}}$ has an increased value relative to time **t9** and Reset_pfm_timer is de-asserted, which results in $V_{\text{timer_ramp}}$ beginning to rise starting at time **t11**. At time **t12**, $V_{\text{timer_ramp}}$ crosses $V_{\text{ref_pfm}}$, resulting in assertion of PFM_timer_comp . Because PFM_timer_comp is asserted before the next rising edge of PWM_CS , offset current adaption will increase the PFM_timer_comp period. At time **t13**, Reset_pfm_timer is asserted, which resets (e.g., grounds) $V_{\text{timer_ramp}}$. Also, $I_{\text{pfm_ramp}}$ is reduced at time **t13** (e.g., by increasing the offset current based on Sel_pfm_timer).

[0070] At time **t14**, $I_{\text{pfm_ramp}}$ has a reduced value relative to time **t11** and Reset_pfm_timer is de-asserted, which results in $V_{\text{timer_ramp}}$ beginning to rise starting at time **t14**. At time **t15**, the next rising edge of PWM_CS is asserted before $V_{\text{timer_ramp}}$ crosses $V_{\text{ref_pfm}}$ (PFM_timer_comp is thus not asserted). Because the next rising edge of PWM_CS occurs before $V_{\text{timer_ramp}}$ crosses $V_{\text{ref_pfm}}$ again, offset current adaption is complete. In other words, iterative toggling of $I_{\text{pfm_ramp}}$ value between two values indicates that the PFM timer interval is sufficiently aligned with PWM_CS . Accordingly, $\text{PFM_adaptation_done}$ is asserted at time **t15**. After time **t15**, the PFM mode begins with the PFM timer interval aligned with PWM_CS .

[0071] FIG. 7 is a diagram **700** showing example inductor current and output voltage resulting from a switching converter controller without an adaptive PFM timer. In the diagram **700**, a PFM mode starts at time **t1**. Before time **t1**, I_L ramps up and down based on PWM mode operations. At

time **t1**, IL ramps up and down based on PFM mode operations, resulting in a higher average output current and an increase in VOUT. At time **t2**, VOUT reaches a maximum of about 1.25 V. At time **t2**, switching is paused until time **t3**, which causes VOUT to fall. At time **t3**, IL ramps up and down based on PFM mode operations causing VOUT to rise from time **t3** to time **t4**. At time **t4**, switching is paused until time **t5**, wherein causes VOUT to fall. At time **t5**, IL ramps up and down based on PFM mode operations causing VOUT to rise from time **t5** to time **t6**. At time **t6**, switching is paused until time **t7**, wherein causes VOUT to fall. At time **t7**, IL ramps up and down based on PFM mode operations causing VOUT to rise from time **t7** to time **t8**. At time **t8**, switching is paused until time **t9**, wherein causes VOUT to fall. At time **t9**, IL ramps up and down based on PFM mode operations causing VOUT to rise from time **t9** to time **t10**. At time **t10**, switching is paused until time **t11**, wherein causes VOUT to fall. At time **t11**, IL ramps up and down based on PFM mode operations causing VOUT to rise from time **t11** to time **t12**. At time **t12**, switching is paused until time **t13**, wherein causes VOUT to fall. At time **t13**, IL ramps up and down based on PFM mode operations causing VOUT to rise from time **t13** to time **t14**. At time **t14**, switching is paused until time **t15**. At time **t15**, IL ramps up and down based on PFM mode operations causing VOUT to rise from time **t15** to time **t16**. At time **t14**, switching is paused. In the example of FIG. 7, VOUT ripple without an adaptive PFM timer has a maximum VOUT ripple of approximately 0.05 V or 50 mV (based on 1.25 V-1.2 V) at time **t2**, which is significant and may be unacceptable for some power management scenarios.

[0072] FIG. 8 is a diagram **800** showing example inductor current and output voltage resulting from a switching converter controller with an adaptive PFM timer (e.g., the adaptive PFM timer circuit **164** in FIG. 1, or the adaptive PFM timer circuit **164A** in FIG. 4). In the diagram **800**, a PFM mode starts at time **t1**. Before time **t1**, IL ramps up and down based on PWM mode operations. At time **t1**, IL ramps up and down based on PFM mode operations, resulting in a higher average output current and an increase in VOUT. At time **t2**, VOUT reaches a maximum of about 1.214 V. Between time **t2** to time **t3**, IL ramps up and down with pause intervals based on PFM mode operations causing VOUT to settle over time to around 1.2 V. In the example of FIG. 8, VOUT ripple with an adaptive PFM timer has a maximum VOUT ripple of approximately 0.014 V or 14 mV at time **t2**, which is 3× improvement relative to the VOUT ripple in FIG. 7.

[0073] FIG. 9 is a flowchart showing an example switching converter controller method **900**. As shown, the switching converter controller method **900** includes identifying a PFM mode trigger at block **902**. In some examples, the PFM mode trigger is a light load condition or low-power mode control signal. At block **904**, adaptation control signals (e.g., Sel_pfm_timer<5:0>, Reset_pfm_timer, V_valley_zero) are provided to a PFM timer circuit (e.g., the adaptive PFM timer circuit **164** in FIG. 1, or the adaptive PFM timer circuit **164A** in FIG. 4) responsive to the PFM mode trigger. At block **906**, adaptation results of the PFM timer circuit (e.g., the timing of PFM_timer_comp relative to PWM_CS) are monitored responsive to the adaptation control signals. At block **908**, a PFM mode signal (e.g., PFM_adaptation_done or resulting signals) is asserted to initiate a PFM mode responsive to the adaptation results indicating PFM timing is ready (e.g., PFM_time_comp is aligned with a target PFM timing such as the PWM switching period). In some examples, PFM timing is ready when PFM_time_comp is aligned with the rising edge of PWM_CS.

[0074] FIG. 10 is a flowchart showing an example PFM timer circuit method **1000**. The PFM timer circuit method **1000** may be performed, for example, by the adaptive PFM timer circuit **164** in FIG. 1, or the adaptive PFM timer circuit **164A** in FIG. 4. As shown, the PFM timer circuit method **1000** includes receiving a digital control signal (e.g., Sel_pfm_timer<5:0>) at block **1002**. At block **1004**, a PFM timer circuit current (I_pfm_ramp in FIG. 4) is adjusted responsive to the digital control signal. At block **1006**, a capacitor (e.g., the capacitor C_pfm in FIG. 4) is charged responsive to the adjusted current. At block **1008**, a charge on the capacitor is compared to a reference voltage (e.g., V_ref_pfm in FIG. 4) to obtain a PFM timer result (e.g., PFM_timer_comp in FIG. 4). At block

1010, the PFM timer result is output (e.g., to mode control logic as described herein). During a transition interval the PFM timer circuit method **1000** may be repeated until the PFM timer result is aligned with a PWM control signal (e.g., PWM_CS) or a best PFM timer result alignment is determined. In some examples, the PFM timer circuit method **1000** may also include: selecting a reference voltage (e.g., V_valley_zero herein) that is equal to the zero valley current in the PWM mode; and setting the PFM timer circuit current based on the reference voltage. In such examples, adaptive PFM timer control is based on a target reference voltage as well as offset current control based on the digital control signal.

[0075] In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0076] Also, in this description, the recitation “based on” means “based at least in part on.”

Therefore, if X is based on Y, then X may be a function of Y and any number of other factors.

[0077] A device “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

[0078] As used herein, the terms “terminal”, “node”, “interconnection”, “pin” and “lead” are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device or other electronics or semiconductor component and/or a conductor.

[0079] A circuit or device described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.

[0080] While the use of particular transistors is described herein, other transistors (or equivalent devices) may be used instead with little or no change to the remaining circuitry. For example, a field-effect transistor (“FET”) such as an NFET or a PFET, a bipolar junction transistor (BJT—e.g., NPN transistor or PNP transistor), an insulated gate bipolar transistor (IGBT), and/or a junction field effect transistor (JFET) may be used in place of or in conjunction with the devices described herein. The transistors may be depletion mode devices, drain-extended devices, enhancement mode devices, natural transistors or other types of device structure transistors. Furthermore, the devices may be implemented in/over a silicon substrate (Si), a silicon carbide substrate (SiC), a gallium nitride substrate (GaN) or a gallium arsenide substrate (GaAs).

[0081] References may be made in the claims to a transistor's control terminal and its first and second terminals. In the context of a FET, the control terminal is the gate, and the first and second terminals are the drain and source. In the context of a BJT, the control terminal is the base, and the first and second terminals are the collector and emitter.

[0082] References herein to a FET being “ON” means that the conduction channel of the FET is

present and drain current may flow through the FET. References herein to a FET being “OFF” means that the conduction channel is not present so drain current does not flow through the FET. An “OFF” FET, however, may have current flowing through the transistor's body-diode.

[0083] Circuits described herein are reconfigurable to include additional or different components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the resistor shown. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in parallel between the same nodes. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series between the same two nodes as the single resistor or capacitor.

[0084] While certain elements of the described examples are included in an integrated circuit and other elements are external to the integrated circuit, in other examples, additional or fewer features may be incorporated into the integrated circuit. In addition, some or all of the features illustrated as being external to the integrated circuit may be included in the integrated circuit and/or some features illustrated as being internal to the integrated circuit may be incorporated outside of the integrated circuit. As used herein, the term “integrated circuit” means one or more circuits that are: (i) incorporated in/over a semiconductor substrate; (ii) incorporated in a single semiconductor package; (iii) incorporated into the same module; and/or (iv) incorporated in/on the same printed circuit board.

[0085] Uses of the phrase “ground” in the foregoing description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within ± 10 percent of that parameter or, if the parameter is zero, a reasonable range of values around zero.

[0086] Modifications are possible in the described examples, and other examples are possible, within the scope of the claims.

Claims

1. A switching converter controller comprising: mode control logic; and a pulse-frequency modulation (PFM) timer circuit coupled to the mode control logic, the PFM timer circuit including: a first terminal; a second terminal; a digital-to-analog converter (DAC) having a first terminal and a second terminal, the first terminal of the DAC coupled to the first terminal of the PFM timer circuit; a current source having a first terminal, a second terminal, and a third terminal, the second terminal of the current source coupled to the second terminal of the DAC; a capacitor having a first terminal and a second terminal, the first terminal of the capacitor coupled to the first terminal of the current source; a comparator having a first terminal, a second terminal, and a third terminal, the first terminal of the comparator coupled to the first terminal of the capacitor, and the third terminal of the comparator coupled to the second terminal of the PFM timer circuit.
2. The switching converter controller of claim 1, wherein the current source is a first current source, the PFM timer circuit includes a second current source having a first terminal, a second terminal, and a third terminal, the third terminal of the second current source coupled to the first terminal of the capacitor.
3. The switching converter controller of claim 2, wherein the PFM timer circuit has a third terminal and a fourth terminal, the PFM timer circuit includes a switch having a first terminal, a second terminal, a third terminal, and a fourth terminal, the first terminal of the switch is at a reference voltage, the second terminal of the switch is coupled to the third terminal of the PFM timer circuit,

the third terminal of the switch coupled to the fourth terminal of the PFM timer circuit, and the fourth terminal of the switch coupled to the second terminal of the second current source.

4. The switching converter controller of claim 3, wherein the switch is a first switch, the PFM timer circuit has a fifth terminal and includes a second switch having a first terminal, a second terminal, and a third terminal, the first terminal of the second switch coupled to the first terminal of the capacitor, and the second terminal of the second switch coupled to the fifth terminal of the PFM timer circuit.

5. The switching converter controller of claim 4, wherein the mode control logic has a first terminal, a second terminal, a third terminal, and a fourth terminal, the first terminal of the mode control logic coupled to the second terminal of the PFM timer circuit, the second terminal of the mode control logic coupled to the first terminal of the PFM timer circuit, the third terminal of the mode control logic is coupled to the fourth terminal of the PFM timer circuit, and the fourth terminal of the mode control logic is coupled to the fifth terminal of the PFM timer circuit.

6. The switching converter controller of claim 5, further comprising an error amplifier having a first terminal, a second terminal, and a third terminal, the third terminal of the error amplifier coupled to the fourth terminal of the PFM timer circuit.

7. The switching converter controller of claim 6, further comprising a voltage-to-current converter, a peak comparator, and a valley comparator, the voltage-to-current converter having a first terminal, a second terminal, and a third terminal, the peak comparator having a first terminal, a second terminal, and a third terminal, the valley comparator having a first terminal, a second terminal, and a third terminal, the first terminal of the voltage-to-current converter coupled to the third terminal of the error amplifier, the second terminal of the voltage-to-current converter coupled to the second terminal of the peak comparator, and the third terminal of the voltage-to-current converter coupled to the first terminal of the valley comparator.

8. The switching converter controller of claim 7, further comprising driver circuitry having a first terminal, a second terminal, a third terminal, and a fourth terminal, wherein the mode control logic has a fifth terminal, a sixth terminal, a seventh terminal, and an eighth terminal, the fifth terminal of the mode control logic coupled to the third terminal of the peak comparator, the third terminal of the valley comparator coupled to the sixth terminal of the mode control logic, the seventh terminal of the mode control logic coupled to the first terminal of the driver circuitry, and the eighth terminal of the mode control logic coupled to the second terminal of the driver circuitry.

9. The switching converter controller of claim 8, wherein the mode control logic is configured to: receive a clock signal, a PFM timer result, a peak compare result, and a valley compare result; adjust a pulse-width modulation (PWM) control signal state and high-impedance control signal state responsive to the clock signal, the PFM timer result, the peak compare result, and the valley compare result; identify a light load trigger; and provide adaptation control signals to the PFM timer circuit responsive to the light load trigger.

10. A switching converter controller comprising: mode control logic; and a pulse-frequency modulation (PFM) timer circuit coupled to the mode control logic and configured to: receive a digital control signal; adjust a current of the PFM timer circuit responsive to the digital control signal; charge a capacitor responsive to the adjusted current; compare a charge on the capacitor to a reference voltage to obtain a PFM timer result; and output the PFM timer result.

11. The switching converter controller of claim 10, wherein the PFM timer circuit is configured to adjust the current of the PFM timer circuit based on the digital control signal, a switch control signal, and a zero valley reference voltage.

12. The switching converter controller of claim 11, wherein the switching control signal is a first switch control signal, and the PFM timer circuit is configured to discharge the capacitor responsive to a second switch control signal.

13. The switching converter controller of claim 12, further comprising mode control logic configured to provide the digital control signal, the first switch control signal, and a second switch

control signal.

14. The switching converter controller of claim 11, further comprising an error amplifier configured to provide an error result responsive to an output voltage and a reference voltage, wherein the PFM timer circuit is configured to: adjust the current based on the error result when the switch control signal has a first state; and adjust the current based on the zero valley reference voltage when the switch control signal has a second state.

15. The switching converter controller of claim 10, further comprising mode control logic configured to: receive a light load signal; provide adaptation control signals to the PFM timer circuit responsive to the light load signal; monitor adaptation results of the PFM timer circuit responsive to the adaptation control signals; and transition to a PFM mode responsive to the adaptation results indicating PFM timing is aligned with a target PFM timing.

16. A system comprising: a power stage; and a switching converter controller coupled to the power stage, the switching converter controller including a PFM timer circuit and configured to: identify a light load trigger; provide adaptation control signals to the PFM timer circuit responsive to the light load trigger; monitor adaptation results of the PFM timer circuit responsive to the adaptation control signals; and transition to a PFM mode responsive to the adaptation results indicating PFM timing is aligned with a target PFM timing.

17. The system of claim 16, wherein the PFM timer circuit is configured to: receive a digital control signal; adjust a current of the PFM timer circuit responsive to the digital control signal; charge a capacitor responsive to the adjust current; compare a charge on the capacitor to a reference voltage to obtain a PFM timer result; and output the PFM timer result.

18. The system of claim 17, wherein the PFM timer circuit is configured to adjust the current of the PFM timer circuit based on the digital control signal and a switch control signal.

19. The system of claim 18, wherein the PFM timer circuit includes a digital-to-analog converter (DAC), a first current source, and a second current source, the first current source is configured to provide a first current responsive to the switch control signal, the DAC is configured to provide an analog signal responsive to the digital control signal, and the second current source is configured to adjust the first current responsive to the analog signal.

20. The system of claim 19, wherein the switching converter controller is configured to control switches of the power stage using a pulse-width modulation (PWM) mode and a pulse-frequency modulation (PFM) mode, the PWM mode having a zero valley current, and the first current source is configured to provide current, during adaptation of the PFM timer circuit, based on a zero valley reference voltage, the zero valley reference voltage based on the zero valley current.
